Jaividhyarthi Vivekanand

AI & Data Science Student | Python Developer

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LinkedIn GitHub Tickbig Portfolio

PROFESSIONAL SUMMARY

Third-year AI & Data Science student with a 9.2/10 CGPA and experience across three internships. Builds machine learning applications, Python/FastAPI backends, and embedded IoT systems. Two first-place hackathon finishes and one runner-up finish. Seeking opportunities in AI, data analytics, and software development.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript | **Development:** FastAPI, REST APIs, Git
AI and Data: Machine Learning, Data Analysis, Time-Series Modeling
Hardware: Embedded Systems and IoT

WORK EXPERIENCE

Data Analytics Intern | iStudio

March - April 2026

Analyzed datasets using Python and Pandas to identify patterns and summarize findings. Created Matplotlib visualizations and delivered a project report documenting the analysis and results.

Data Analyst Intern | Codtec Solutions

May 2025

Built Python data-analysis deliverables and created visualizations to communicate findings clearly.

Web Development Intern | Elevate Labs

May 2025

Developed frontend features for web applications and deployed updates as part of internship deliverables. Recognised among the Top 10 of 800 interns.

EDUCATION

B.Tech in Artificial Intelligence and Data Science

SRM Valliammai Engineering College | August 2024 - May 2028 (expected)

Currently pursuing third year; CGPA: **9.2/10**.

Higher Secondary Certificate (Class XII)
Sindhi High School, Bangalore

Secondary School Leaving Certificate (Class X)
Jaya Public School

PUBLISHED RESEARCH

- **A-QAOA** - Co-authored research paper published with Springer Nature.
- **Cognitive Mirror AI** - Co-authored research paper published at IEEE NCRTCI 2026.

SOFTWARE AND APPLIED AI PROJECTS

Skyglow | Repository Intelligence Platform

Personal Project | Python, FastAPI, REST APIs, SQLite, Render

Developed and deployed a full-stack platform analyzing live GitHub repository data to produce explainable indicators and report history. Validated backend functionality with **9 passing automated tests**. Live application: skyglow-intelligence.onrender.com.

SIFT | Security Intelligence Filtering and Triage

Team NULLSPACE | NVD, CISA KEV, FIRST EPSS, CVE Data, Explainable Ranking

Contributed to vulnerability triage matching advisories to an organisation’s technology, exposure, and service importance. Produces five prioritised actions with transparent scores, uncertain-version verification flags, and exclusion reasons.

Double SIFT | Structural Integrity and Flow Tracking

Team NULLSPACE | AI Agents, Instruction Provenance, Source Attribution, Plan-Drift Detection

Contributed to a structural defence prototype against indirect prompt injection. Demonstrated an AI email-agent firewall that tracks instruction sources and goal deviations, with trust-based allow, confirm, and block decisions and concise explanations.

ClassLens / TARA | Teaching Analytics for Reflective Advancement

SRM Valliammai Engineering College | Python, FastAPI, Sarvam ASR, MediaPipe PoseLandmarker

Developed a multimodal classroom-recording prototype combining transcription, voice activity detection, speaker diarization, prosody analysis, and pose tracking. Integrated time-based teaching metrics and session reports for evidence-based feedback and academic demonstrations.

Vivian | Digital Personal Assistant

Personal Project | React, Python, FastAPI, SQLite, REST APIs | In development

Building personal and professional workspaces for tasks, notes, messages, events, settings, and approvals. Implemented a database-backed dashboard and assistant APIs, AI chat integration, and database backup support.

SAHARA | Smart Assistant for Humanitarian Aid, Rescue and Action

Nexconn Build Challenge | React, Vite, FastAPI, Neon PostgreSQL, Nexconn Chat and Calling SDKs

Developing humanitarian coordination through help requests, real-time messaging, and in-app calls. Established services and database tables for users, requests, messages, and events. In-app calling works; background incoming-call notifications remain under refinement.

GhostPing | Predictive Lag and Thermal Monitoring

iQOO Hyderabad City Battle | Python, GRU, TensorFlow Lite, Device Telemetry

Designed an on-device monitoring prototype using short telemetry windows and time-series modeling to anticipate network lag and thermal spikes, identifying emerging gaming-performance risks.

RESEARCH AND EMBEDDED SYSTEMS PROJECTS

GridPulse AI | Explainable Multi-Agent Microgrid Orchestration

SRM Valliammai Engineering College | Multi-Agent Systems, Forecasting, Microgrid Simulation

Contributed to a research prototype coordinating renewable forecasting, demand forecasting, fault detection, and battery management. Developed a simulation-based operator dashboard for inspecting grid decisions and explanations. Validation and comparative evaluation are ongoing.

DravidaLM | Offline Language Models for South Indian Languages

Team NULLSPACE | Python, Transformers, Tokenization, Corpus Processing, NVIDIA Jetson AGX

Researching Tamil, Telugu, Kannada, and Malayalam models with dialect coverage, code-mixed text, and offline edge deployment. Conducted initial experiments using approximately **24 million tokens**; further corpus preparation, implementation, and evaluation are ongoing.

ATRP | Adaptive Trust Routing Protocol for IoT Networks

Academic Research | IoT Simulation, Trust-Based Routing, Graph Algorithms

Investigated adaptive routing using node reliability and dynamic path costs. Benchmarked against Dijkstra and random-walk strategies and prepared a research manuscript submitted for review.

AQUAWAVE | Adaptive Sonar Transmitter for AUVs

Team NULLSPACE | STM32F407, DSP, LFM Chirps, Phase-Coded Waveforms, SD Logging

Designed a sonar-transmitter payload for autonomous underwater vehicles, including waveform generation, embedded control, logging, and safety considerations. Developed the architecture and waveform approach as a foundation for hardware implementation and validation.

Motor/Pump Predictive Health and Dry-Run Protection

Personal Project | ESP32, ACS712 Current Sensor, Python

Built an embedded system analyzing motor current signatures for dry-run conditions and indicators of early wear. Converted sensor readings into motor-health indicators and integrated automatic protective cut-off in response to detected faults.

Phantom | Autonomous Phantom Load Detection Device

SRM Valliammai Engineering College | Microcontroller, Current Sensing, Relay Control

Developed embedded hardware that identifies standby electrical loads from live current measurements and isolates connected appliances through relay control, supporting reduced unnecessary electricity consumption.

Q.Farm | Intelligent Crop Advisory

Academic / Hackathon Project | Python, Random Forest, Quantum-Inspired Scoring

Developed a crop-advisory prototype combining Random Forest predictions with quantum-inspired scoring. Integrated predictive modeling and recommendation scoring to explore data-driven crop selection.

AWARDS AND ACHIEVEMENTS

- **1st Prize, Neurobots National Hackathon 2026, Thrissur.** Winning team; team prize valued at INR 1,00,000.
- **1st Prize, Erode Sengunthar Engineering College Hackathon 2026.**
- **2nd Prize, NEXORA National Hackathon 2026, Coimbatore.** Cybersecurity Track.
- **Smart India Hackathon 2026 - Internal Qualifier.** Selected through the college internal round with Team NULLSPACE, presenting AQUAWAVE.
- **1st Runner-Up, IT Buzzfeed 2024.** Competed individually against 47 teams.
- **Top 10 of 800 interns, Elevate Labs.** Recognition during the Web Development internship.
- **Model United Nations - Honourable Delegate Award.** Represented Norway in UNEA, 8-9 April 2025.

COURSES AND CERTIFICATIONS

Advanced Python Programming
Coursera / upSKILL

Java Programming for Beginners
Coursera / upSKILL

Web Development - Value Added Course
SRM Valliammai Engineering College

MERN Stack Development - Value Added Course
SRM Valliammai Engineering College

Introduction to Cloud Computing
SkillUp

Introduction to Generative AI: Art of the Possible
Amazon Web Services (AWS) | 2025

Generative AI Engineering: Foundations, RAG & Deployment
HCL GUVI

The AI Skills You Need to Stay Relevant in the Next 2 Years
WsCube Tech

Artificial Intelligence and Machine Learning Training
Skill Ladders × Kshitij 2026, IIT Kharagpur

Financial Technology Job Simulation
JPMorgan Chase & Co. - Forage

GenAI Powered Data Analytics Job Simulation
Tata - Forage | 2026

INTERNSHIP COMPLETION CERTIFICATES

Web Development - Elevate Labs (2025)
Data Science / Data Analytics - Codtec Solutions (2025)
Data Analytics - iStudio (2026)